
Nathan Seidle
Joel Bartlett
Rob Reynolds

Combos in 45
minutes or less!*

*Totes Guaranteed

2002

Credit: Me



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Way more versatility than just radio, the bladeRF x 40.

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SparkFun micro:climate kit

 KIT-14217
~~\$114.95~~ \$54.95



SparkFun micro:arcade kit

 KIT-14218
~~\$49.95~~ \$46.78



SparkFun moto:bit

 DEV-14213
~~\$14.95~~ \$11.18



SparkFun weather:bit

 DEV-14214
~~\$14.95~~ \$10.95

[LATEST BLOG POST](#)

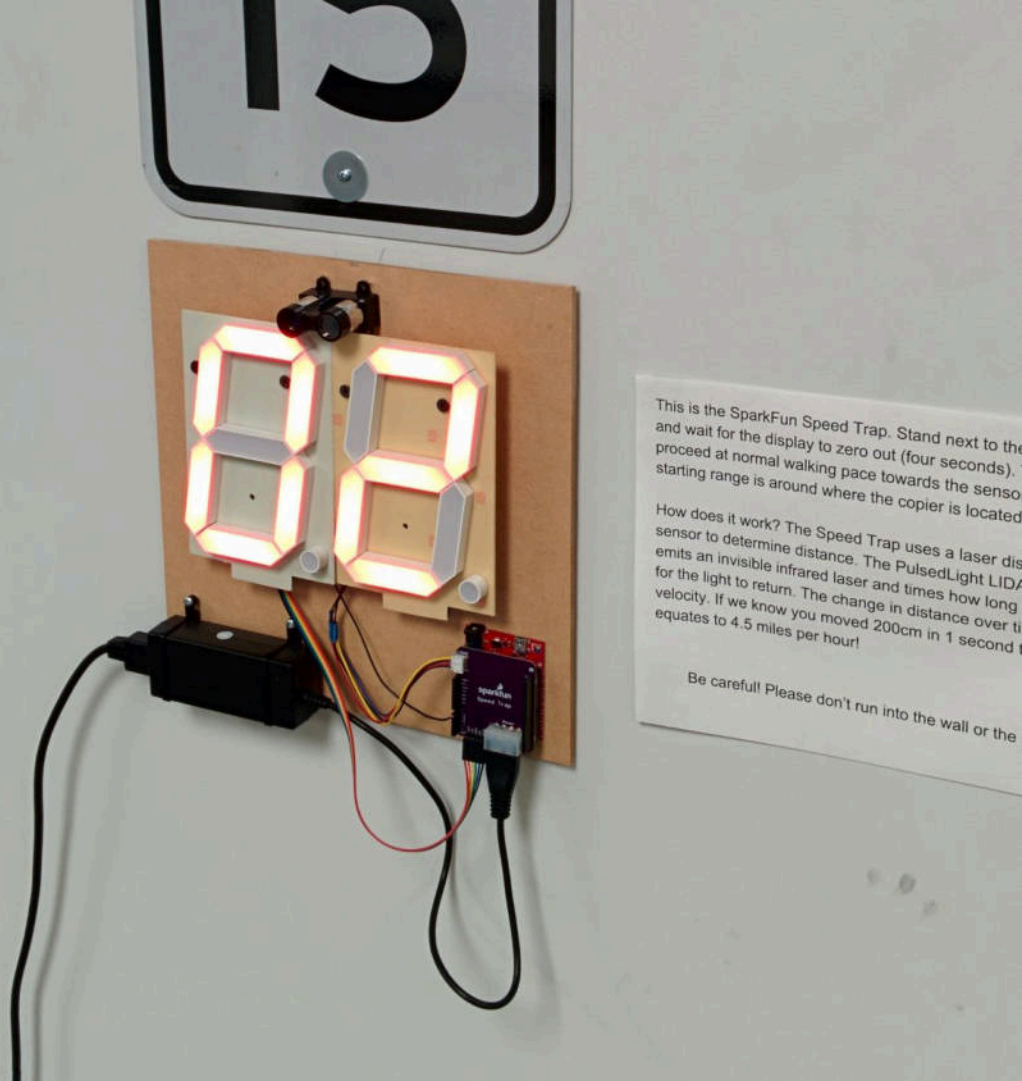
Desk of an Engineer: the Mary Edition



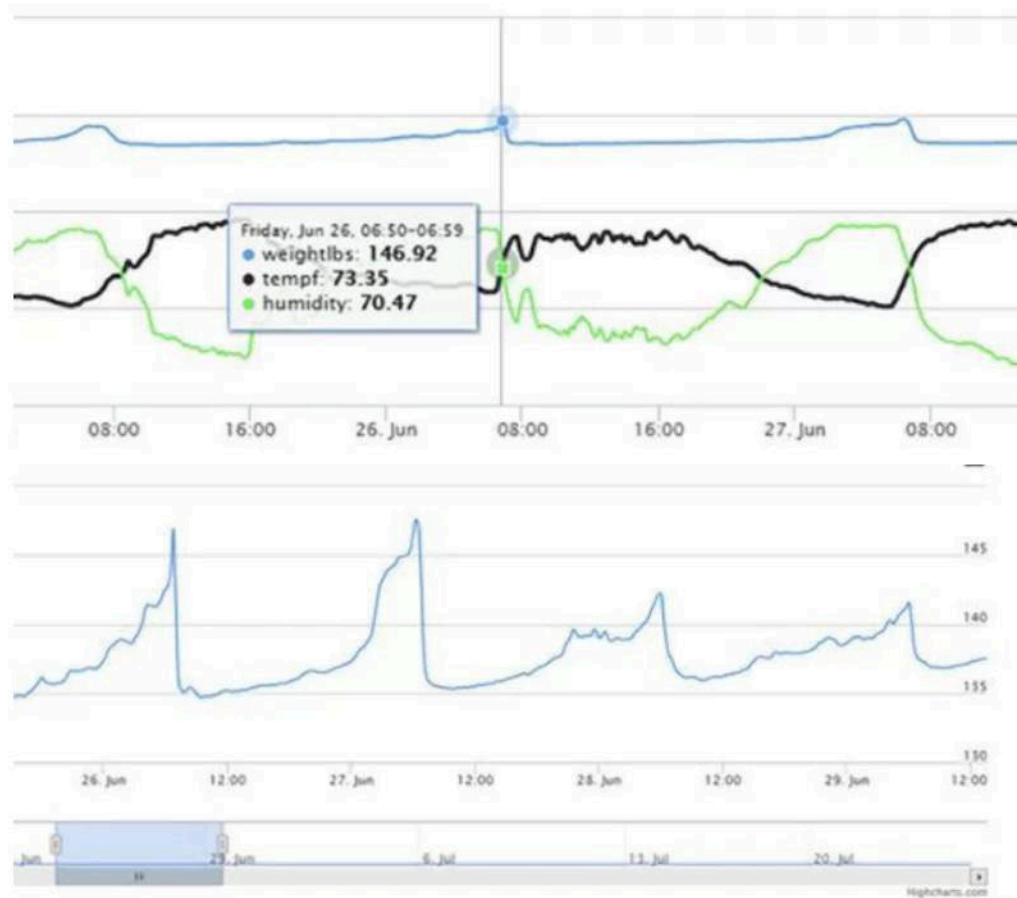
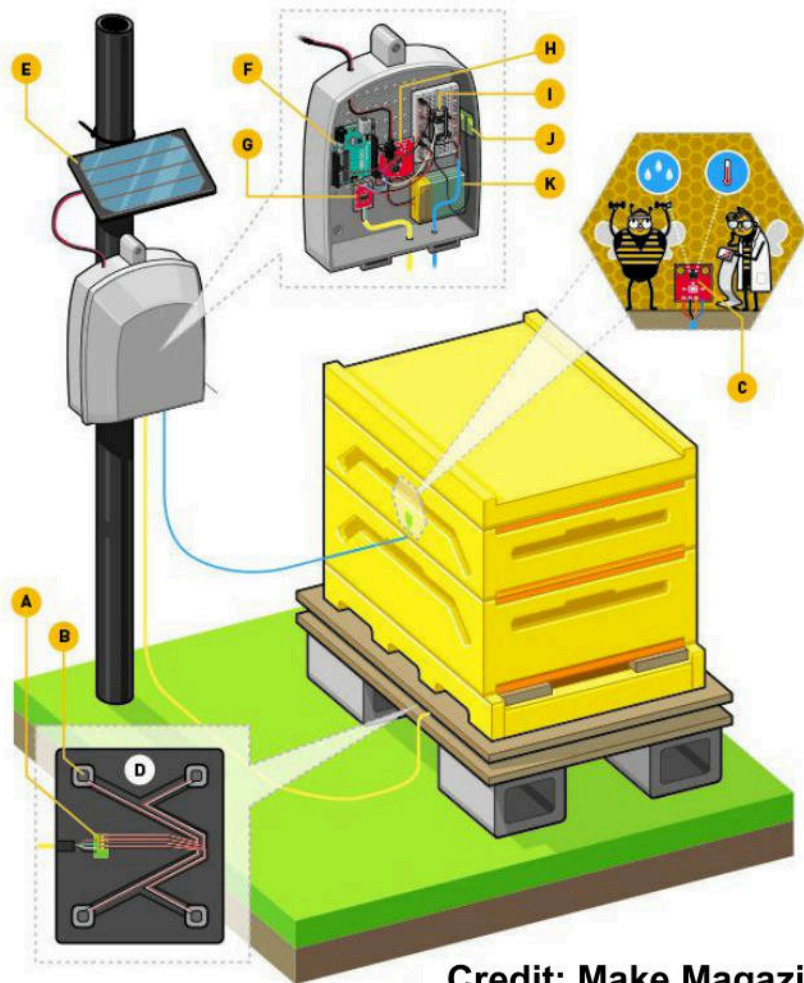
FREE SHIPPING



Credit: Benjamin Rasmussen







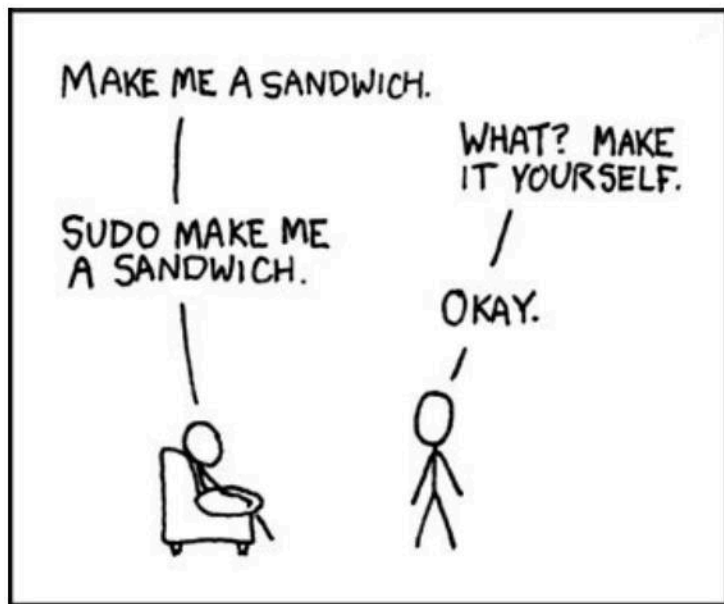
Credit: Make Magazine



Credit: SparkFun



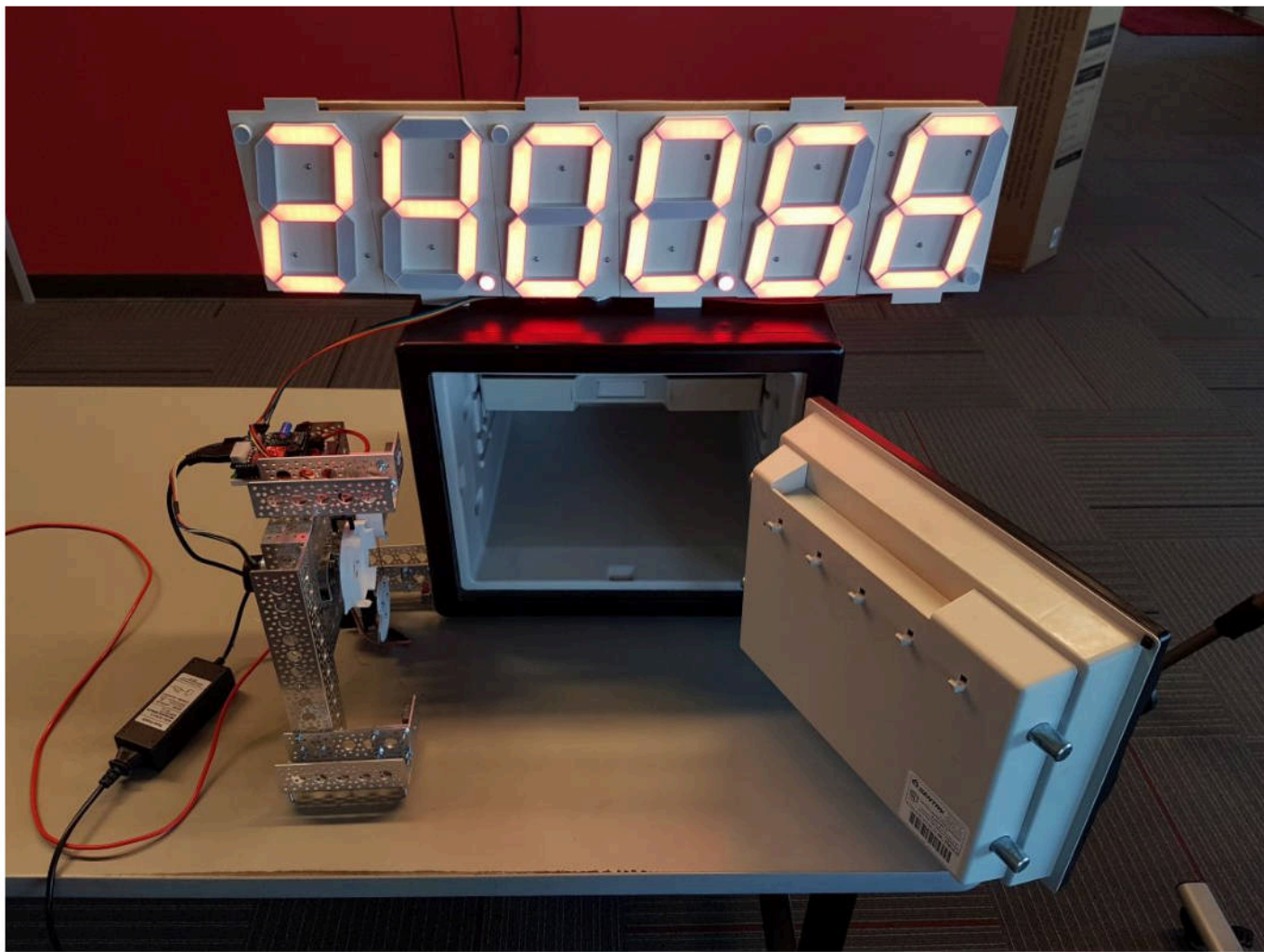
Credit: SparkFun

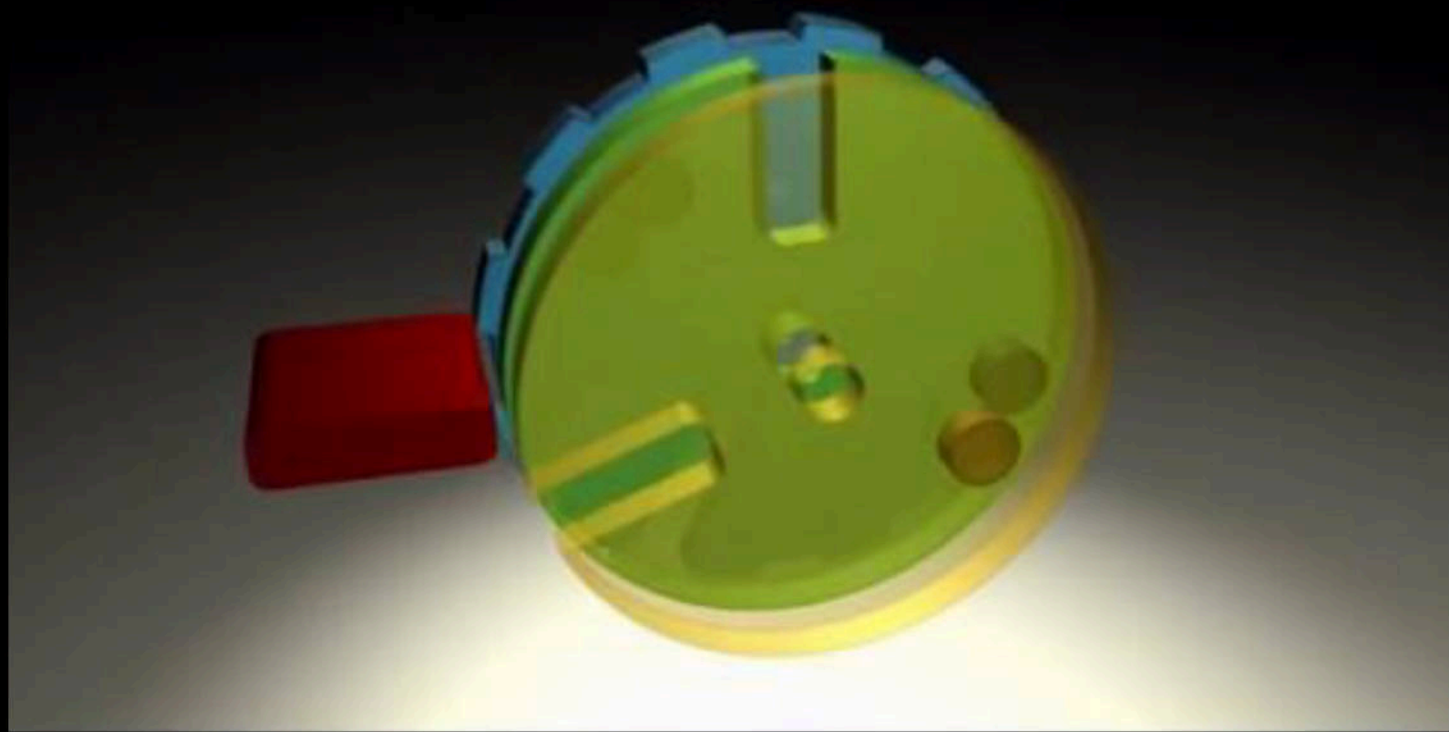


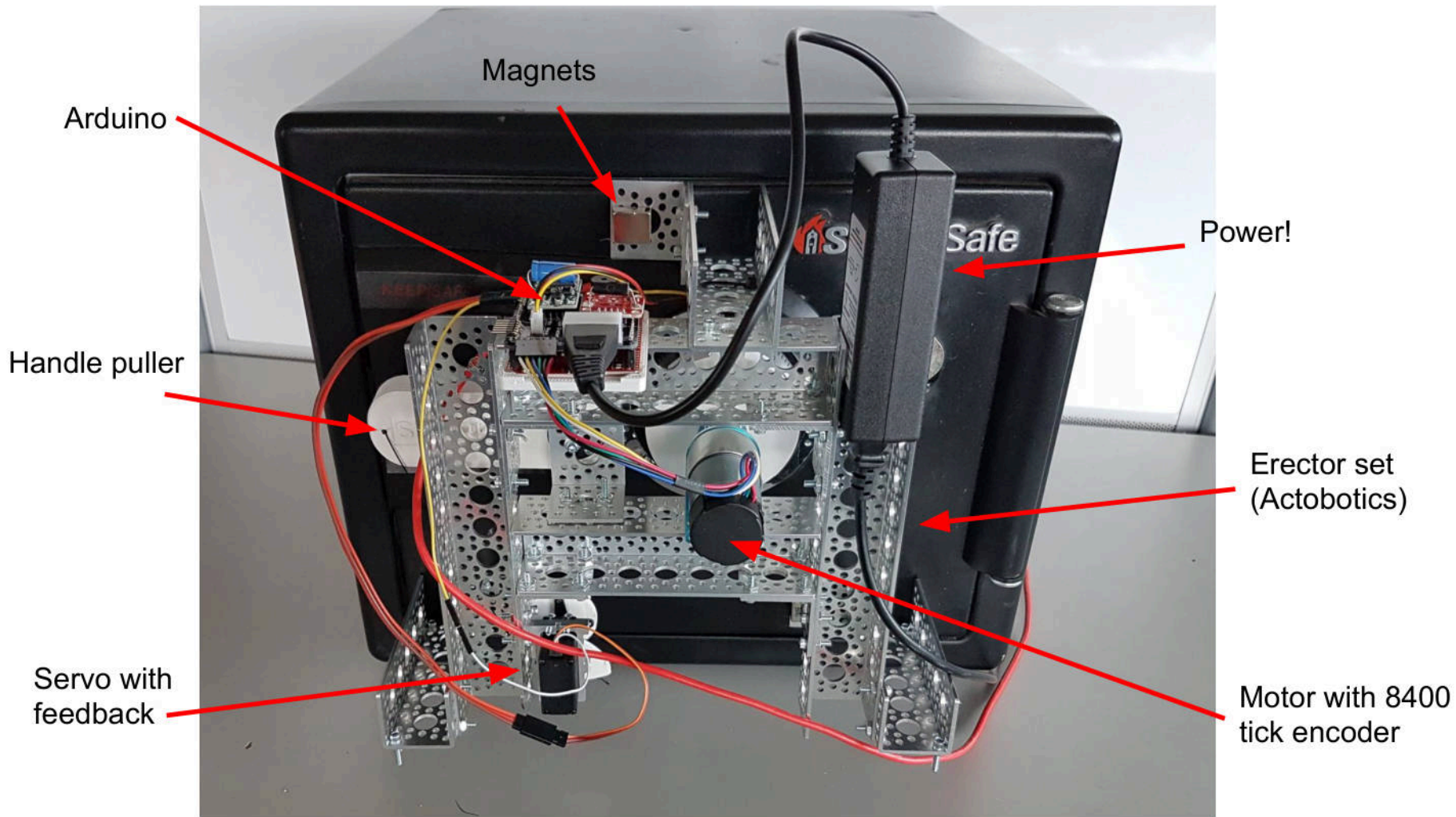
Credit: xkcd



Credit: SentrySafe / Liberty Safes







Arduino

Handle puller

Servo with feedback

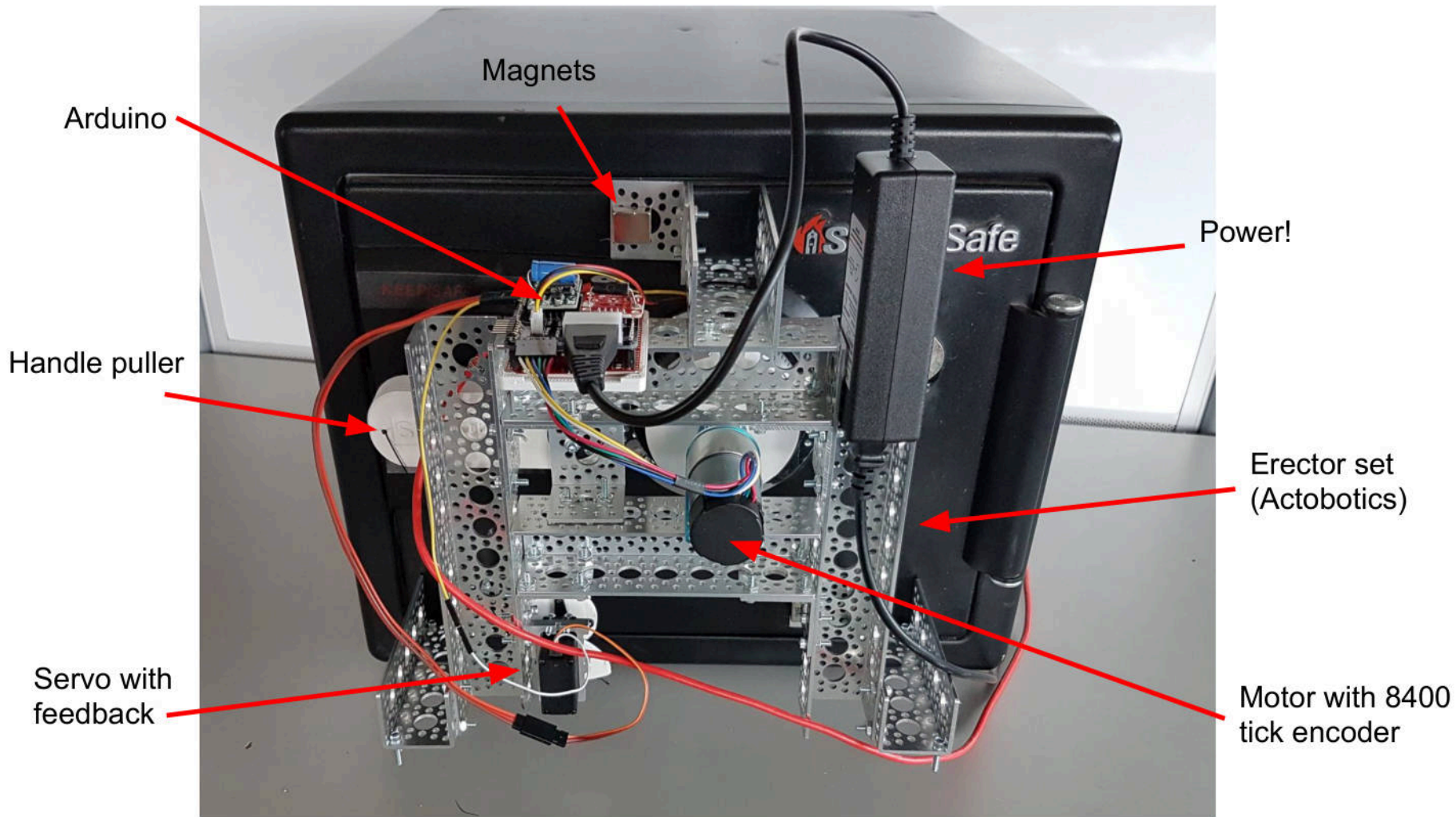


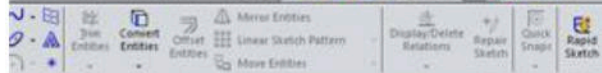
Power!

Erector set (Actobotics)

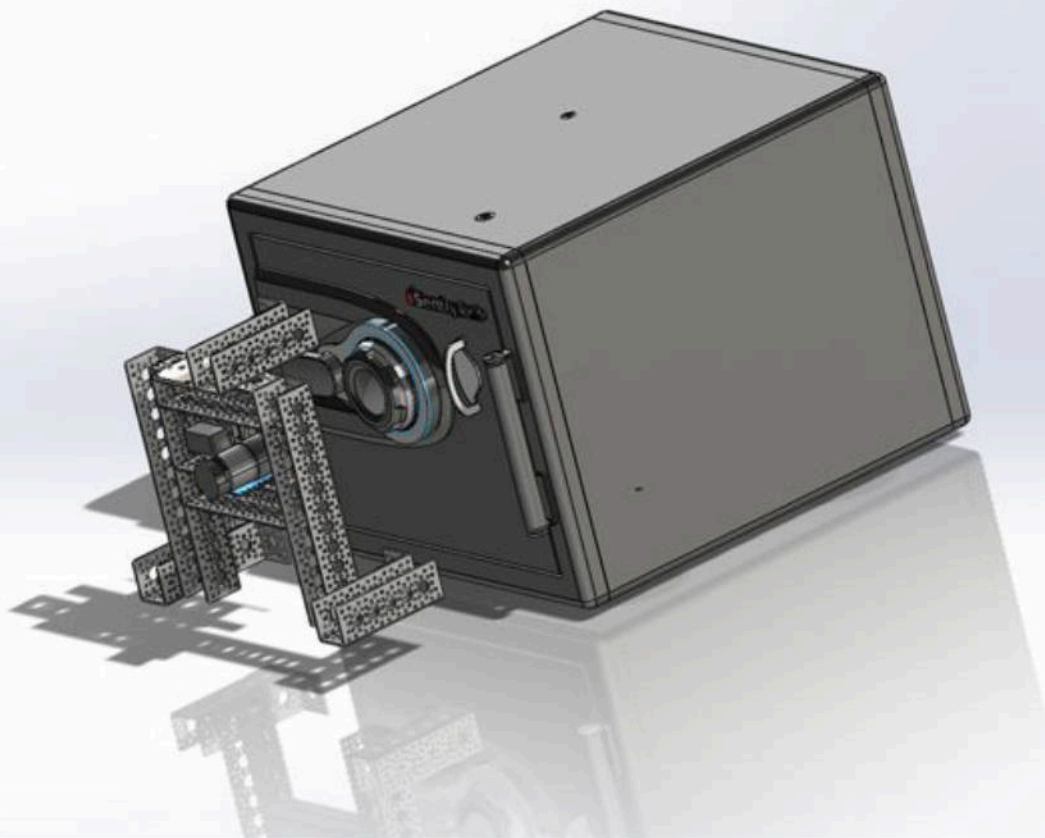
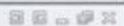
Motor with 8400 tick encoder

Credit: Pololu

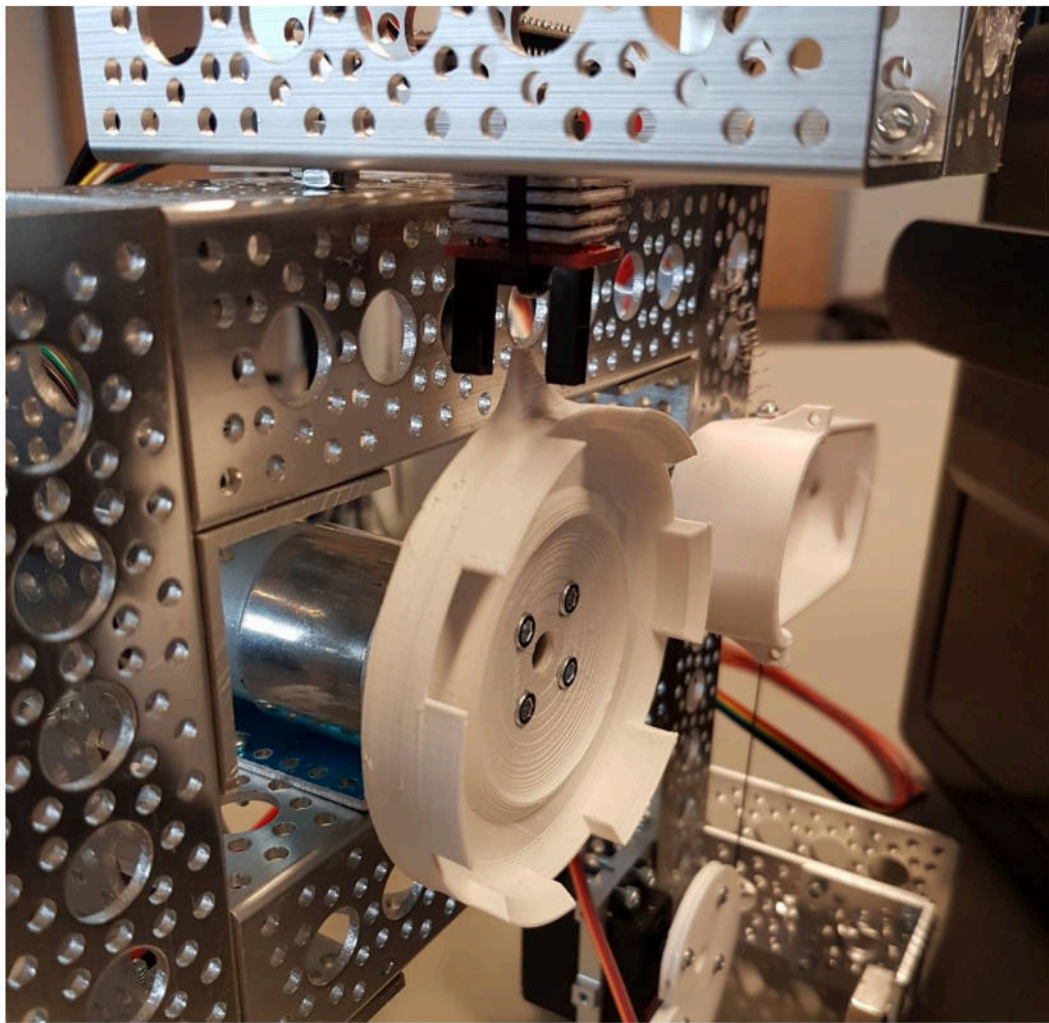


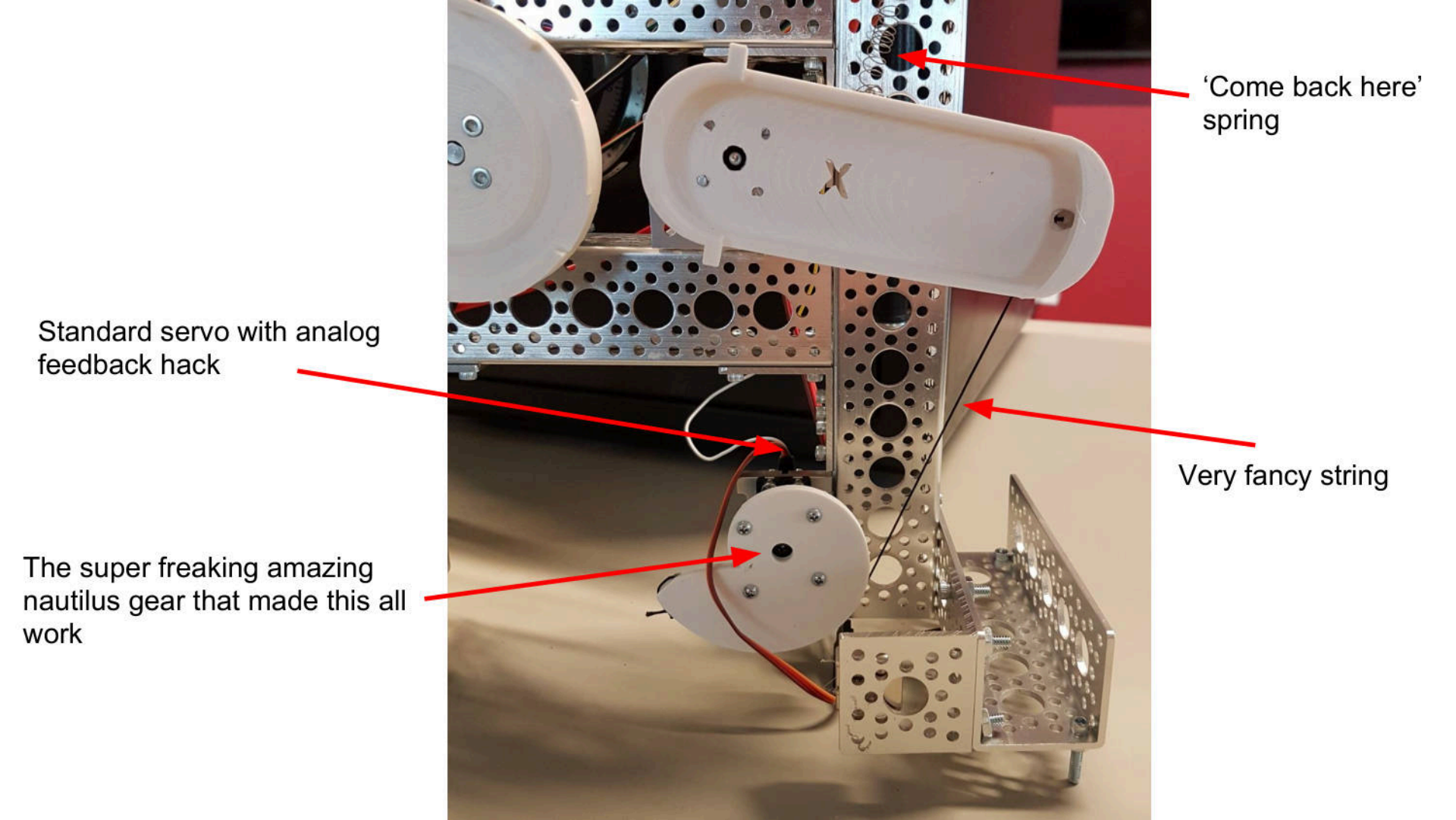


multigert Office Products



9.1.1



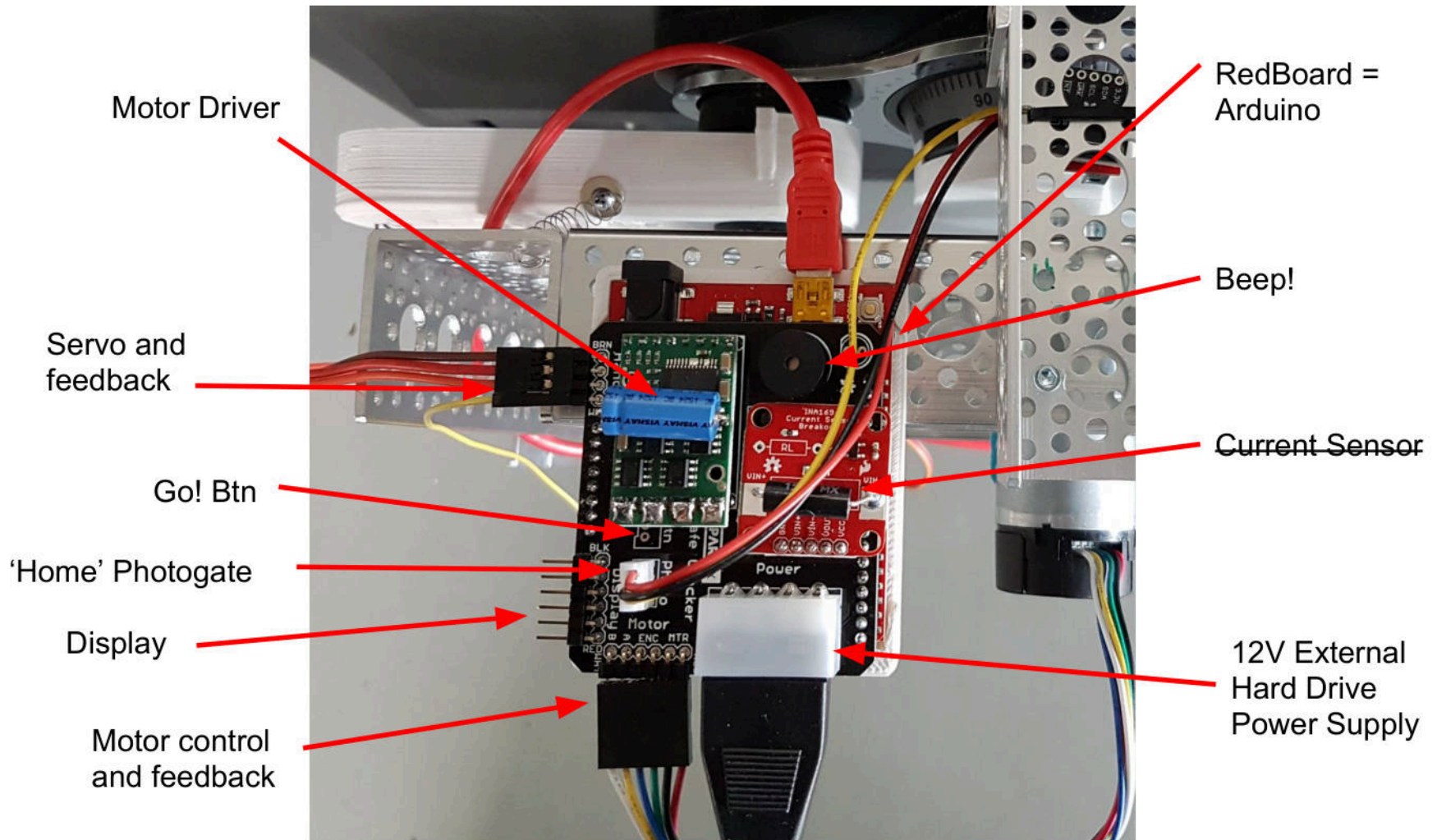


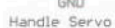
'Come back here'
spring

Standard servo with analog
feedback hack

Very fancy string

The super freaking amazing
nautilus gear that made this all
work





Problem Domain:

100^3 combinations

10 seconds per test

115 days (worst case)

Combinations:
 100^3 combinations

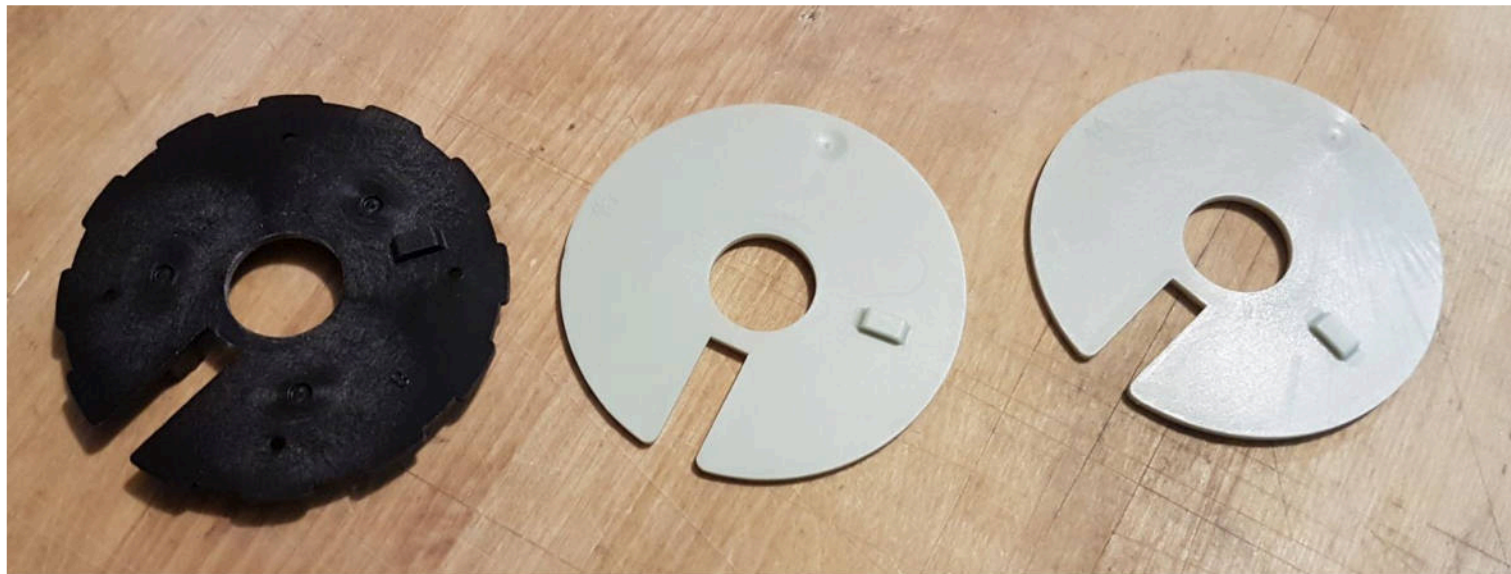
Exploits

Combinations:

~~**100^3 combinations**~~

33^3 combinations = 4.15 days

Exploits



Exploits

Combinations:

~~**100^3 combinations**~~

~~**33^3 combinations = 4.15 days**~~

Disc C has 12 indents

$33^2 * 12 = 1.5$ days

Exploits



Disc C:

Outer diameter: 2.815" (71.5mm)

Width of solution slot: 0.239"

Width of 11 indents: 0.249" +/- 0.002"

8.84" (Circumference) / 8400 ticks

0.001" / tick

~10 ticks smaller

Exploits

Combinations:

~~**100^3 combinations**~~

~~**33^3 combinations = 4.15 days**~~

~~**Disc C has 12 indents**~~

~~**$33^2 * 12 = 1.5$ days**~~

Disc C has a skinny indent

$33^2 * 1 = 3$ hours

Exploits





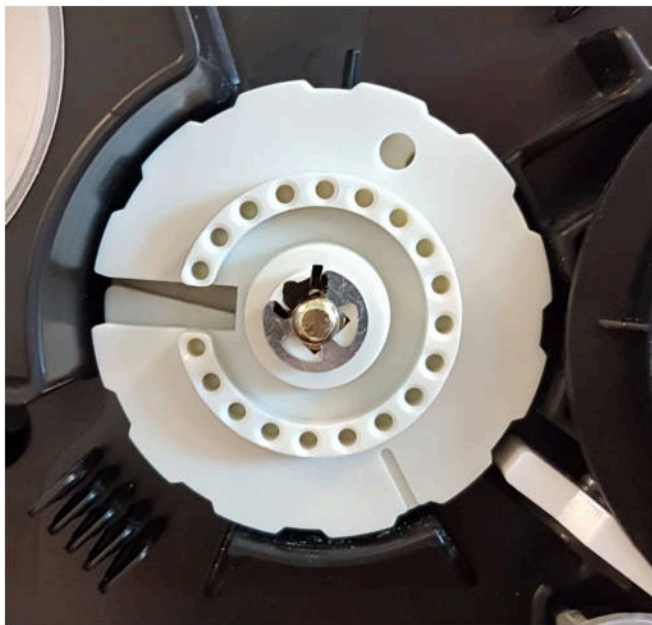












'New' Disc C:

Outer diameter: 2.456" (62.4mm)

Width of solution slot: 0.250"

Width of 11 indents: 0.201" +/- 0.002"

7.72" (Circumference) / 8400 ticks

0.00092" / tick

**~54 ticks LARGER
(5 times easier to hack)**

Exploits



File Edit Setup Control Window Help

6) Set starting combos

7) Calibrate handle servo

8) Test handle button

9) Test indent centers

s) Start cracking

How many measurements would you like to take? (Start with 1)1

Test 1 of 1. IndentNum[0] Encoder[8049] / Dial[96] / Width[474] / Depth[0]

Test 1 of 1. IndentNum[1] Encoder[354] / Dial[4] / Width[485] / Depth[0]

Test 1 of 1. IndentNum[2] Encoder[1051] / Dial[13] / Width[468] / Depth[0]

Test 1 of 1. IndentNum[3] Encoder[1758] / Dial[21] / Width[503] / Depth[0]

Test 1 of 1. IndentNum[4] Encoder[2458] / Dial[29] / Width[478] / Depth[0]

Test 1 of 1. IndentNum[5] Encoder[3153] / Dial[38] / Width[496] / Depth[0]

Test 1 of 1. IndentNum[6] Encoder[3848] / Dial[46] / Width[535] / Depth[0]

Test 1 of 1. IndentNum[7] Encoder[4534] / Dial[54] / Width[496] / Depth[0]

Test 1 of 1. IndentNum[8] Encoder[5248] / Dial[62] / Width[495] / Depth[0]

Test 1 of 1. IndentNum[9] Encoder[5943] / Dial[71] / Width[490] / Depth[0]

Test 1 of 1. IndentNum[10] Encoder[6638] / Dial[79] / Width[484] / Depth[0]

Test 1 of 1. IndentNum[11] Encoder[7344] / Dial[87] / Width[481] / Depth[0]

Largest indent number: 6

Largest indent is now set to test

Indent locations recorded to EEPROM

Press key to continue

6) Set starting combos

7) Calibrate handle servo

8) Test handle button

9) Test indent centers

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Combinations:

~~**100^3 combinations**~~

~~**33^3 combinations = 4.15 days**~~

~~**Disc C has 12 indents**~~

~~**$33^2 * 12 = 1.5$ days**~~

Disc C has a large indent

$33^2 * 1 = 3$ hours

Exploits

Test Time:

Resetting Dials = 10s / test

Exploits

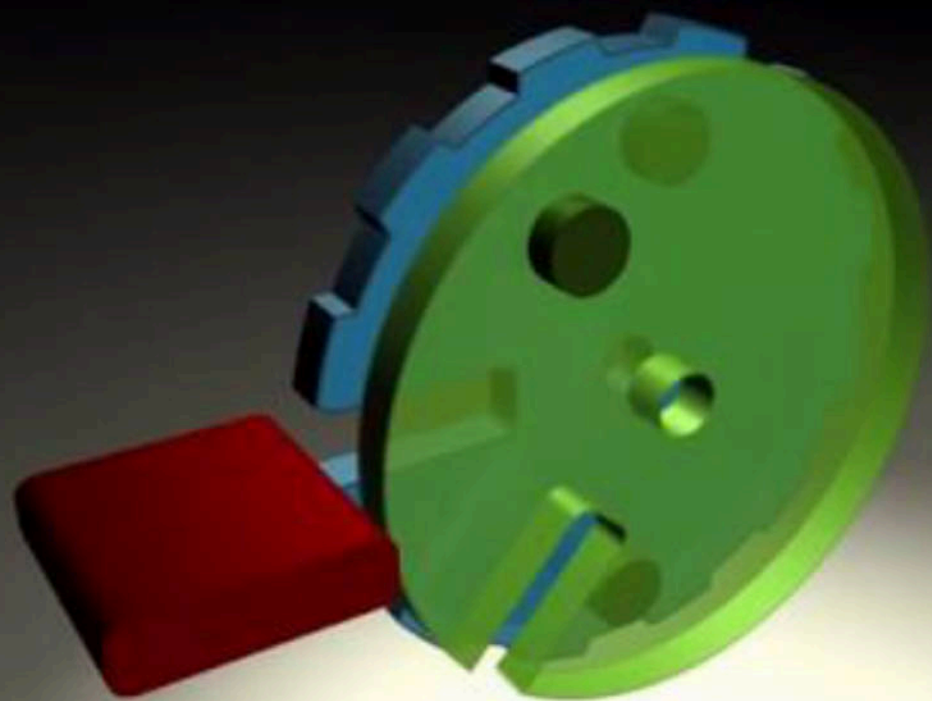
Test Time:

~~**Resetting Dials = 10s / test**~~

'Set testing' = 4s / test

1.2 hours

Exploits





Test Time:

~~Resetting Dials = 10s / test~~

~~'Set testing' = 4s / test~~

~~1.2 hours~~

45 minutes!

~~Exploits~~ **Luck**

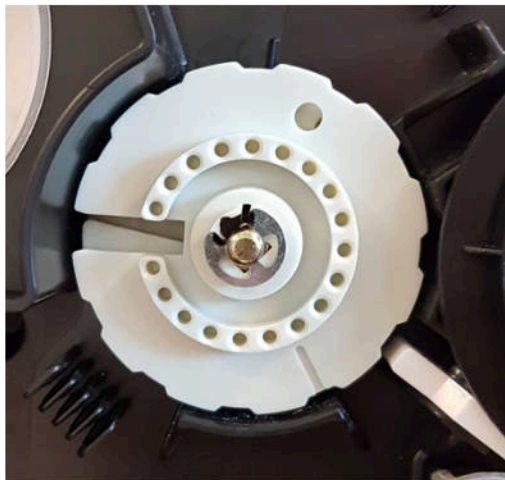
**How do I protect
myself!?**



Credit: starwarsblog.starwars.com



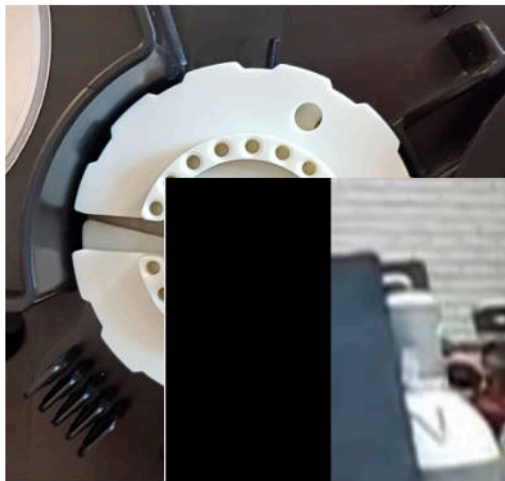
One of these is not like the others...



“The S&G 6730 ... has only a $\pm .5$ dialing tolerance, essentially giving a 1 digit window to hit. While many locksmiths might prefer the S&G 6730, it can be notoriously difficult to open and very unforgiving to human error. In addition, slight alterations to the lock (for example, if the dial or the dial ring was bumped during shipping) can shift the combination, rendering the lock unusable.”



-Hayman Safes: Lock Ratings



5 dialing
window to
fer the S&G
o open and
dition, slight
f the dial or
opping) can
he lock

gs

Future Research

Measuring complete

Smallest to largest width [Width]/[Depth]

Indent 8: [1911] / [1130]

Indent 1: [1925] / [1122]

Indent 3: [1953] / [1091]

Indent 0: [1955] / [1099]

Indent 11: [1966] / [1105]

Indent 2: [1992] / [1100]

Indent 9: [1994] / [1126]

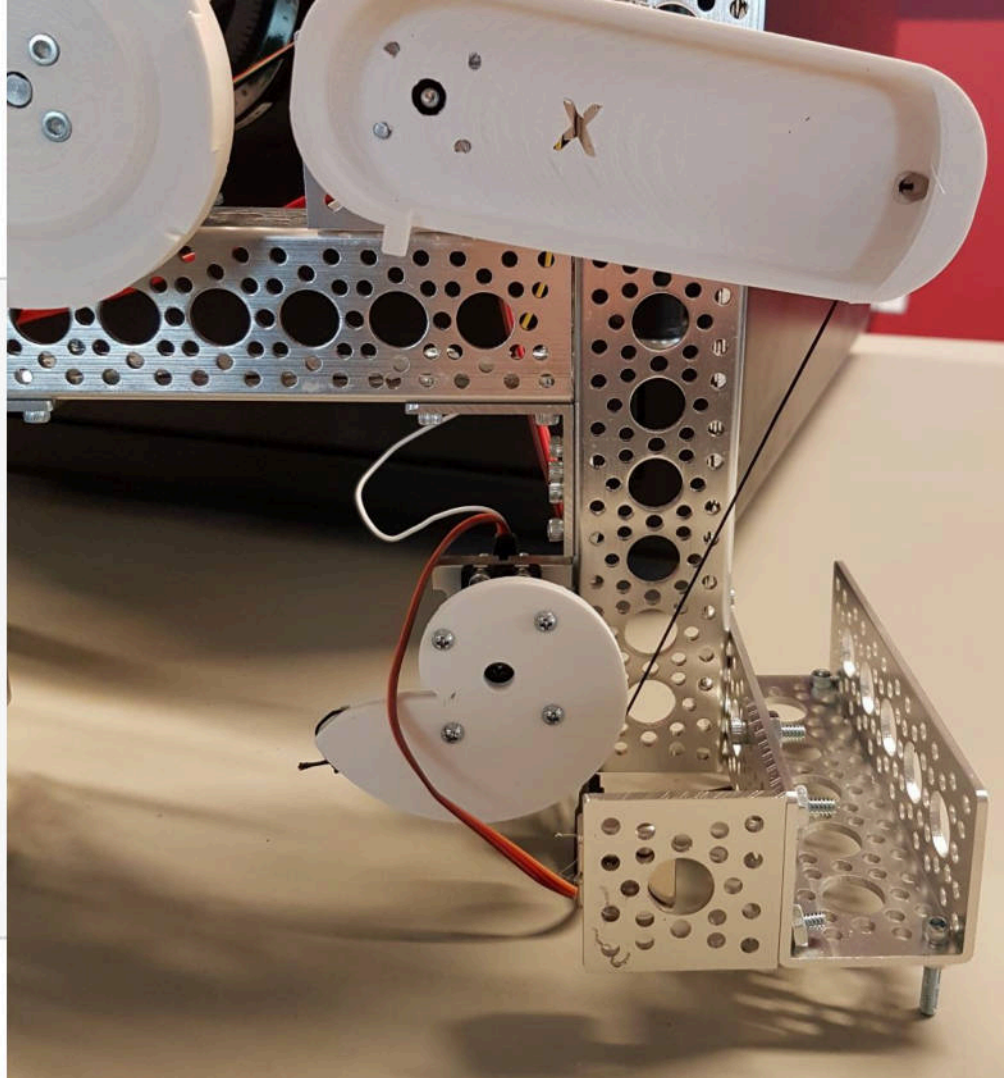
Indent 7: [2011] / [1098]

Indent 10: [2036] / [1096]

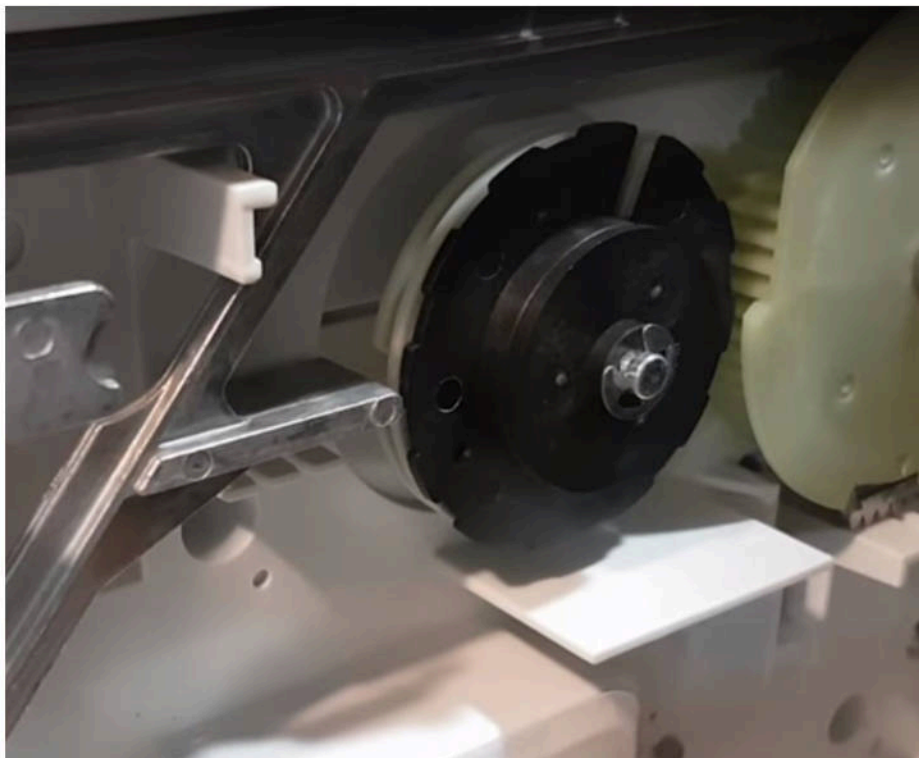
Indent 4: [2077] / [1109]

Indent 5: [2083] / [1100]

Indent 6: [2114] / [1096]



Future Research



Future Research



Credit: iRobot

Future Research



Is it open yet?



nathan@sparkfun.com



Demo fail!